

Smell Like Home: An Olfactory-Driven VR Experience through a Scent-Camera Interface

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Abstract

Smell Like Home is a multi-sensory virtual reality experience that explores how olfactory memory shapes one's sense of home and belonging. Blending scent, emotion, and immersive spatial design, the work considers how smell, an intensely emotional yet rarely represented sense in digital media, can awaken memory and transform the perception of space. Participants move through virtual environments reconstructed from the artist's memories, guided by a scent camera that links digital imagery to real scent. Each photograph or interaction activates a custom scent diffuser mounted on the VR controller, releasing corresponding scents through Wi-Fi in sync with the player's interaction.

The project examines the subtle and profound bond between smell and memory, revealing how a single scent can traverse time and summon long-buried emotions and scenes. By combining sensory design with technical precision, *Smell Like Home* does more than enhance immersion through scent; it reconstructs the sense of presence that smell evokes. It reflects on how immersive technology can revive the sensory language of memory, turning scent into a bridge between past and present, through which memories can be momentarily lived again.

Index Terms: Olfactory, Multi-sensory Experience, Virtual Reality, Cultural Memory, Embodied Interaction

1 Description

1.1 Concept

Smell Like Home is a multi-sensory virtual reality experience that explores the relationship between scent, memory, and the meaning of home. Participants navigate a sequence of intimate spaces reconstructed from the artist's memories, where smell becomes the primary trigger for emotional recall. Through olfactory interaction and spatial storytelling, the work reflects on how immersive technology can reawaken embodied memory and reconstruct a sense of belonging through the senses.

1.2 Narrative Structure

The narrative unfolds within environments reconstructed from the artist's memories, each reflecting a different stage in the evolving definition of home. Everyday objects dispersed across these spaces serve as anchors of memory, carrying both emotional and sensory resonance. When participants photograph these objects using a handheld scent-camera, each image triggers a short narrative fragment and a corresponding scent, revealing part of a personal story. As

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Figure 1: Participant interacting with the scent camera during the *Smell Like Home* VR experience. The custom diffuser releases scents in sync with virtual actions, linking olfactory cues to storytelling.

the experience progresses, these fragments connect to form a larger emotional portrait of home.

The storytelling structure draws inspiration from the video game *What Remains of Edith Finch* [1], in which independent vignettes gradually converge into a unified emotional arc. There are no textual instructions or interface prompts; instead, participants are guided by curiosity and the intuitive desire to discover which objects might release a scent or unlock a memory.

The experience follows a reverse temporal structure. It begins with an onboarding scene that introduces the scent camera and core interaction. The following three scenes correspond to different life stages, each presenting a distinct interpretation of home shaped by memory and scent. The experience concludes in a virtual gallery where all captured photographs are displayed together, offering a reflective space to revisit the scents, stories, and emotions.

1.3 Interaction and Interface

Prior research demonstrates that olfactory stimuli can be precisely controlled, rapidly switched, and synchronized with user interaction or audiovisual content, with user studies confirming accurate scent recognition and reliable experiential feedback [2]. Building on this foundation, *Smell Like Home* employs a handheld scent camera as both a tangible interface and a narrative device tied strongly to the core mechanic. Physically grasping the camera grounds the act of photographing in a tactile gesture, strengthening immersion through embodied interaction. Conceptually, the camera mediates between remembering and revealing, linking image, scent, and story. Each photograph not only records a visual trace but also triggers a custom scent diffuser mounted on the VR controller, releasing synchronized olfactory cues via Wi-Fi communication. Through this integration, photography becomes an act of sensory recollection. Technical details are discussed in the *Implementation* section.

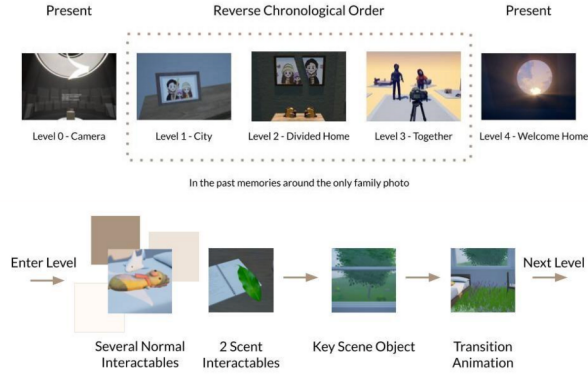


Figure 2: Narrative structure of *Smell Like Home*. The experience unfolds through five scenes arranged along a reverse timeline. Each memory scene links a specific scent to a stage in the evolving understanding of home.

1.4 Cultural Dimension

The selection of scents carries both personal and cultural significance. Lychee, coffee, and osmanthus each represent a layer of the artist's understanding of home, rooted in the urban and seasonal sensory landscape of the artist's hometown. Prior research shows that olfactory cues can evoke emotionally rich, place-based memories that contribute to identity formation [3, 4, 5]. Drawing on perspectives from The Multi-sensory Museum [6], this work approaches space as an embodied, multi-sensory experience rather than a purely visual construct.

Through the intersection of personal recollection and sensory design, *Smell Like Home* becomes both an autobiographical work and a reflection on how smell encodes cultural memory, transforming ephemeral sensations into a language of presence.

2 Fit with the Scope of IEEE VR 2026 'XR Gallery'

Aligned with the 'XR Gallery' theme, *(Re)Imagine: Bridging the Past, Present and Future through Immersive Experiences*, this work bridges temporal and emotional dimensions through olfactory memory. By positioning smell as the central narrative medium, it reframes immersive storytelling as an embodied experience that connects technology with human emotion.

In contemporary digital culture, immersion is often dominated by vision and sound. *Smell Like Home* re-centers the body as a perceptual site, emphasizing smell as a carrier of presence and emotional continuity. Rather than reproducing reality, the work uses scent to make memory tangible, revealing how immersive technology can reshape perceptions of time, identity, and belonging.

3 Implementation

Smell Like Home is developed using **Meta Quest 2** and **Unreal Engine 5**, alongside a custom scent diffuser driven by an **Arduino Nano 33 IoT** micro-controller. The diffuser is mounted on the VR controller and contains three scent chambers, each associated with a core memory scene. Miniature air pumps regulate diffusion intensity, while Wi-Fi communication synchronizes scent release with user interaction in real time.

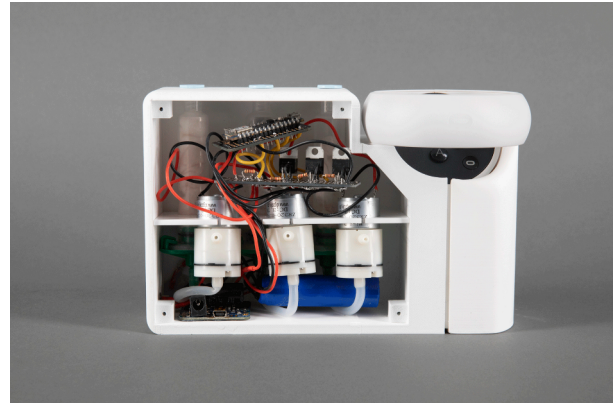


Figure 3: Custom scent diffuser used in *Smell Like Home*. The device integrates three scent chambers, micro air pumps, and an Arduino Nano 33 IoT board for wireless communication with the virtual environment.

When a participant captures a photograph in VR, Unreal Engine triggers the corresponding scent with minimal latency, preserving the illusion of immediate physical response. The experience requires a standing play area of approximately **3 × 3 meters**, allowing participants to turn and move lightly within the environment. Spatial sound and environmental cues support orientation without explicit navigation interfaces. Each scene encourages slow, reflective exploration, transforming photography into an embodied act of remembering.

For exhibition, the experience is single-user setup, with each session lasting approximately 13 minutes.

4 Conclusion

Smell Like Home explores olfaction as a central narrative medium in virtual reality, integrating scent, embodied interaction, and spatial memory through a scent-camera interface. The work demonstrates how immersive technologies can engage memory through sensory activation rather than visual representation alone. By foregrounding smell as a carrier of presence and meaning, the project contributes an alternative approach to multisensory storytelling within XR.

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